

SQL PRACTICE NOTES

TOPIC: DDL, DML, DCL, TCL COMMANDS

SECTION 1: DDL COMMANDS

1. CREATE TABLE

```
MySQL 8.0 Command Line Cli x + v
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> mysql -u root -p
-> CREATE DATABASE bank_db;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'mysql -u root -p CREATE DATABASE bank_db' at line 1
mysql> ^C
mysql> CREATE DATABASE bank_db;
Query OK, 1 row affected (0.01 sec)

mysql> SHOW DATABASES;
+-----+
| Database |
+-----+
| bank_db |
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
5 rows in set (0.02 sec)

mysql> USE bank_db;
Database changed
mysql> CREATE TABLE bank_transactions (
-> txn_id INT PRIMARY KEY,
-> customer_name VARCHAR(50),
-> branch_name VARCHAR(50),
-> transaction_type VARCHAR(20),
```

```
MySQL 8.0 Command Line Cli x + v - □ x

mysql> USE bank_db;
Database changed
mysql> CREATE TABLE bank_transactions (
    -> txn_id INT PRIMARY KEY,
    -> customer_name VARCHAR(50),
    -> branch_name VARCHAR(50),
    -> transaction_type VARCHAR(20),
    -> amount DECIMAL(10,2),
    -> transaction_date DATE
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.01 sec)

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
--+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
--+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
```

```
MySQL 8.0 Command Line Cli x + v - □ x

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.01 sec)

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
--+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
--+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
--+
6 rows in set (0.01 sec)

mysql>
```

2. ALTER TABLE ADD COLUMN

6 rows in set (0.01 sec)

```
mysql> ALTER TABLE bank_transactions
-> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> DESCRIBE bank_transactions;
```

Field	Type	Null	Key	Default	Extra
txn_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
branch_name	varchar(50)	YES		NULL	
transaction_type	varchar(20)	YES		NULL	
amount	decimal(10,2)	YES		NULL	
transaction_date	date	YES		NULL	
account_no	varchar(20)	YES		NULL	

7 rows in set (0.00 sec)

3. MODIFY COLUMN

```
mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
--+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
--+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(100)  | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
--+
7 rows in set (0.01 sec)

mysql>
```

4. RENAME TABLE

	branch_name		varchar(50)		YES	
	transaction_type		varchar(20)		YES	
	amount		decimal(10,2)		YES	
	transaction_date		date		YES	
	account_no		varchar(20)		YES	

+-----+-----+-----+-----+-----+-----+

--+

7 rows in set (0.01 sec)

```
mysql> RENAME TABLE bank_transactions
      -> TO customer_transactions;
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> SHOW TABLES;
```

+-----+
Tables_in_bank_db
+-----+
customer_transactions
+-----+

1 row in set (0.00 sec)

```
mysql> RENAME TABLE customer_transactions
      -> TO bank_transactions;
```

Query OK, 0 rows affected (0.02 sec)

```
mysql>
```

5. TRUNCATE TABLE

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

|
| transaction_date | date          | YES | | NULL |
|
| account_no       | varchar(20)   | YES | | NULL |
|
+-----+-----+-----+-----+-----+
--+
7 rows in set (0.01 sec)

mysql> RENAME TABLE bank_transactions
      -> TO customer_transactions;
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.00 sec)

mysql> RENAME TABLE customer_transactions
      -> TO bank_transactions;
Query OK, 0 rows affected (0.02 sec)

mysql> TRUNCATE TABLE bank_transactions;
Query OK, 0 rows affected (0.09 sec)

mysql> SELECT * FROM bank_transactions;
Empty set (0.00 sec)

mysql> |
```

6. DROP TABLE

```
MySQL 8.0 Command Line Cli × + v - □ ×

mysql> RENAME TABLE bank_transactions
-> TO customer_transactions;
Query OK, 0 rows affected (0.03 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| customer_transactions |
+-----+
1 row in set (0.00 sec)

mysql> RENAME TABLE customer_transactions
-> TO bank_transactions;
Query OK, 0 rows affected (0.02 sec)

mysql> TRUNCATE TABLE bank_transactions;
Query OK, 0 rows affected (0.09 sec)

mysql> SELECT * FROM bank_transactions;
Empty set (0.00 sec)

mysql> CREATE TABLE bank_backup(
-> id INT
-> );
Query OK, 0 rows affected (0.04 sec)

mysql> DROP TABLE bank_backup;
Query OK, 0 rows affected (0.03 sec)

mysql>
```

SECTION 2: DML COMMANDS



```
mysql> DROP TABLE bank_backup;  
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO bank_transactions  
-> (txn_id, customer_name, branch_name, transaction_type, amount  
, transaction_date)  
-> VALUES  
-> (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),  
-> (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),  
-> (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),  
-> (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),  
-> (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),  
  
-> (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),  
-> (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'  
' ),  
-> (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),  
-> (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),  
-> (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');  
Query OK, 10 rows affected (0.01 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM bank_transactions;
```

```
+-----+-----+-----+-----+-----+  
+-----+-----+-----+-----+  
| txn_id | customer_name | branch_name | transaction_type | amou  
nt      | transaction_date | account_no |  
+-----+-----+-----+-----+-----+  
+-----+-----+-----+-----+  
|      101 | Ravi          | Hyderabad  | Deposit          | 500  
0.00    | 2024-01-05    | NULL       |  
+-----+-----+-----+-----+-----+
```

```
MySQL 8.0 Command Line Cli × + v - □ ×

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 500 |
0.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 200 |
0.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 1200 |
0.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 800 |
0.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 350 |
0.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 1500 |
0.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 100 |
0.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 900 |
0.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 400 |
0.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 1100 |
0.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

INSERT

```
MySQL 8.0 Command Line Cli × + v - □ ×

10 rows in set (0.00 sec)

mysql> INSERT INTO bank_transactions
-> INSERT INTO bank_transactions
-> (txn_id,customer_name,branch_name,transaction_type,amount
,transaction_date)
-> VALUES
-> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
ERROR 1064 (42000): You have an error in your SQL syntax; check
the manual that corresponds to your MySQL server version for the
right syntax to use near 'INSERT INTO bank_transactions
(txn_id,customer_name,branch_name,transaction_type' at line 2
mysql> INSERT INTO bank_transactions
-> (txn_id,customer_name,branch_name,transaction_type,amount
,transaction_date)
-> VALUES
-> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 500 |
0.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 200 |
0.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 1200 |
0.00 | 2024-01-08 | NULL |
```

```
MySQL 8.0 Command Line Cli × + ▾
mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 500 |
0.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 200 |
0.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 1200 |
0.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 800 |
0.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 350 |
0.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 1500 |
0.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 100 |
0.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 900 |
0.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 400 |
0.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 1100 |
0.00 | 2024-01-16 | NULL |
| 111 | Venu | Vizag | Deposit | 700 |
0.00 | 2024-01-18 | NULL |
+-----+-----+-----+-----+-----+
11 rows in set (0.00 sec)
```

UPDATE

```
MySQL 8.0 Command Line Cli x + v - □ x

| 110 | Madhu | Vijayawada | Deposit | 1100
0.00 | 2024-01-16 | NULL |
| 111 | Venu | Vizag | Deposit | 700
0.00 | 2024-01-18 | NULL |
+-----+-----+-----+-----+-----+
-----+-----+-----+-----+
11 rows in set (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
-> SET amount=5000
-> WHERE txn_id=105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT
-> ^C
mysql> SELECT * FROM bank_transactions
-> WHERE txn_id=105;
+-----+-----+-----+-----+-----+
-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amou
nt | transaction_date | account_no |
+-----+-----+-----+-----+-----+
-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000
.00 | 2024-01-11 | NULL |
+-----+-----+-----+-----+-----+
-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

DELETE

```
-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> DELETE FROM bank_transactions
-> WHERE txn_id=111;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM bank_transactions;
```

```
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
|-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 500 |
0.00 | 2024-01-05 | NULL | | |
| 102 | Sita | Hyderabad | Withdrawal | 200 |
0.00 | 2024-01-06 | NULL | | |
| 103 | Kiran | Vijayawada | Deposit | 1200 |
0.00 | 2024-01-08 | NULL | | |
| 104 | Anil | Vizag | Deposit | 800 |
0.00 | 2024-01-10 | NULL | | |
| 105 | Priya | Hyderabad | Withdrawal | 500 |
0.00 | 2024-01-11 | NULL | | |
| 106 | Ramesh | Vizag | Deposit | 1500 |
0.00 | 2024-01-12 | NULL | | |
| 107 | Keerthi | Vijayawada | Withdrawal | 100 |
0.00 | 2024-01-13 | NULL | | |
| 108 | Rahul | Hyderabad | Deposit | 900 |
0.00 | 2024-01-14 | NULL | | |
| 109 | Sneha | Vizag | Withdrawal | 400 |
0.00 | 2024-01-15 | NULL | | |
```

```
MySQL 8.0 Command Line Cli × + ▾
mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 500 |
0.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 200 |
0.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 1200 |
0.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 800 |
0.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 500 |
0.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 1500 |
0.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 100 |
0.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 900 |
0.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 400 |
0.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 1100 |
0.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

SELECT

```
MySQL 8.0 Command Line Cli × + v - □ ×

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 500 |
0.00 | 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 200 |
0.00 | 2024-01-06 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 1200 |
0.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 800 |
0.00 | 2024-01-10 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 500 |
0.00 | 2024-01-11 | NULL |
| 106 | Ramesh | Vizag | Deposit | 1500 |
0.00 | 2024-01-12 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 100 |
0.00 | 2024-01-13 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 900 |
0.00 | 2024-01-14 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 400 |
0.00 | 2024-01-15 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 1100 |
0.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

WHERE

```
MySQL 8.0 Command Line Cli × + v - □ ×

| 110 | Madhu | Vijayawada | Deposit | 1100
0.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+
-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT *
      -> FROM bank_transactions
      -> WHERE transaction_type='Deposit';
+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amou
nt | transaction_date | account_no |
+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 500
0.00 | 2024-01-05 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 1200
0.00 | 2024-01-08 | NULL |
| 104 | Anil | Vizag | Deposit | 800
0.00 | 2024-01-10 | NULL |
| 106 | Ramesh | Vizag | Deposit | 1500
0.00 | 2024-01-12 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 900
0.00 | 2024-01-14 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 1100
0.00 | 2024-01-16 | NULL |
+-----+-----+-----+-----+
-----+-----+
6 rows in set (0.00 sec)

mysql>
```

ORDER BY

```
MySQL 8.0 Command Line Cli × + ▾

mysql> SELECT *
      -> FROM bank_transactions
      -> ORDER BY amount DESC;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 106 | Ramesh | Vizag | Deposit | 15000.00 |
| 2024-01-12 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 |
| 2024-01-08 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 |
| 2024-01-16 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 |
| 2024-01-14 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 |
| 2024-01-10 | NULL |
| 101 | Ravi | Hyderabad | Deposit | 5000.00 |
| 2024-01-05 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 |
| 2024-01-11 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 |
| 2024-01-15 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 |
| 2024-01-06 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 |
| 2024-01-13 | NULL |
+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

SECTION 3: DCL COMMANDS

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

+-----+-----+-----+-----+-----+
+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
+-----+
| 106 | Ramesh | Vizag | Deposit | 15000.00 |
| 2024-01-12 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 |
| 2024-01-08 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 |
| 2024-01-16 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 |
| 2024-01-14 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 |
| 2024-01-10 | NULL |
| 101 | Ravi | Hyderabad | Deposit | 5000.00 |
| 2024-01-05 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 |
| 2024-01-11 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 |
| 2024-01-15 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 |
| 2024-01-06 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 |
| 2024-01-13 | NULL |
+-----+-----+-----+-----+-----+
+-----+
10 rows in set (0.00 sec)

mysql> CREATE USER 'Auditor1' '@'localhost'
-> IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.03 sec)

mysql>
```

GRANT

```
MySQL 8.0 Command Line Cli x + v - □ x

+-----+-----+
| 106 | Ramesh | Vizag | Deposit | 15000.00 |
| 2024-01-12 | NULL | | | |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 |
| 2024-01-08 | NULL | | | |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 |
| 2024-01-16 | NULL | | | |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 |
| 2024-01-14 | NULL | | | |
| 104 | Anil | Vizag | Deposit | 8000.00 |
| 2024-01-10 | NULL | | | |
| 101 | Ravi | Hyderabad | Deposit | 5000.00 |
| 2024-01-05 | NULL | | | |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 |
| 2024-01-11 | NULL | | | |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 |
| 2024-01-15 | NULL | | | |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 |
| 2024-01-06 | NULL | | | |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 |
| 2024-01-13 | NULL | | | |
+-----+-----+
+-----+-----+
10 rows in set (0.00 sec)

mysql> CREATE USER 'Auditor1'@'localhost'
-> IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT
-> ON bank_db.bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Show permissions

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×
| 109 | Sneha | Vizag | Withdrawal | 4000.00
| 2024-01-15 | NULL | | |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00
| 2024-01-06 | NULL | | |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00
| 2024-01-13 | NULL | | |
+-----+-----+-----+-----+-----+
+-----+
10 rows in set (0.00 sec)

mysql> CREATE USER 'Auditor1'@'localhost'
-> IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT
-> ON bank_db.bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
+-----+
-----+
| Grants for Auditor1@localhost
|
+-----+
-----+
| GRANT USAGE ON *.* TO 'Auditor1'@'localhost'
|
| GRANT SELECT ON `bank_db`.`bank_transactions` TO 'Auditor1'@'localh
ost' |
+-----+
-----+
2 rows in set (0.00 sec)

mysql>
```

REVOKE

```
MySQL 8.0 Command Line Cli x + v
| 2024-01-13 | NULL |
+-----+-----+-----+-----+
+-----+
10 rows in set (0.00 sec)

mysql> CREATE USER 'Auditor1' '@'localhost'
-> IDENTIFIED BY 'pass123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT SELECT
-> ON bank_db.bank_transactions
-> TO 'Auditor1' '@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1' '@'localhost';
+-----+
-----+
| Grants for Auditor1@localhost
|
+-----+
-----+
| GRANT USAGE ON *.* TO 'Auditor1' '@'localhost`
|
| GRANT SELECT ON `bank_db`.`bank_transactions` TO `Auditor1` '@'localh
ost` |
+-----+
-----+
2 rows in set (0.00 sec)

mysql> REVOKE SELECT
-> ON bank_db.bank_transactions
-> FROM 'Auditor1' '@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

SECTION 4: TCL COMMANDS

```
MySQL 8.0 Command Line Cli × + ▾
    -> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
+-----+
| Grants for Auditor1@localhost |
+-----+
| GRANT USAGE ON *.* TO 'Auditor1'@'localhost' |
| GRANT SELECT ON `bank_db`.`bank_transactions` TO 'Auditor1'@'localhost' |
+-----+
2 rows in set (0.00 sec)

mysql> REVOKE SELECT
    -> ON bank_db.bank_transactions
    -> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SET autocommit=0;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @@autocommit;
+-----+
| @@autocommit |
+-----+
| 0 |
+-----+
1 row in set (0.00 sec)

mysql>
```

COMMIT

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

mysql> SET autocommit=0;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @@autocommit;
+-----+
| @@autocommit |
+-----+
|             0 |
+-----+
1 row in set (0.00 sec)

mysql> UPDATE bank_transactions
-> SET amount=6000
-> WHERE txn_id=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM bank_transactions
-> WHERE txn_id=101;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
|      101 | Ravi          | Hyderabad  | Deposit          | 6000.00 |
| 2024-01-05 | NULL          |             |                  |         |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

SAVEPOINT

```
MySQL 8.0 Command Line Cli
mysql> SELECT @@autocommit;
+-----+
| @@autocommit |
+-----+
| 0 |
+-----+
1 row in set (0.00 sec)

mysql> UPDATE bank_transactions
-> SET amount=6000
-> WHERE txn_id=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM bank_transactions
-> WHERE txn_id=101;
+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 6000.00 | 2024-01-05 | NULL |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

Incorrect Update

Query OK, 0 rows affected (0.00 sec)

mysql> SELECT @@autocommit;

@@autocommit
0

1 row in set (0.00 sec)

mysql> UPDATE bank_transactions

-> SET amount=6000

-> WHERE txn_id=101;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM bank_transactions

-> WHERE txn_id=101;

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Ravi	Hyderabad	Deposit	6000.00	2024-01-05	NULL

1 row in set (0.00 sec)

mysql> COMMIT;

Query OK, 0 rows affected (0.01 sec)

mysql> SAVEPOINT Before_Update;

Query OK, 0 rows affected (0.00 sec)

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 6000.00 |
| 2024-01-05 | NULL | | | |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
-> SET amount=99999
-> WHERE txn_id=102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM bank_transactions
-> WHERE txn_id=102;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no | | | |
+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 99999.00 |
| 2024-01-06 | NULL | | | |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

Rollback

```
MySQL 8.0 Command Line Cli × + - □ ×

Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM bank_transactions
      -> WHERE txn_id=102;
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 99999.00 |
| 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT * FROM bank_transactions
      -> WHERE txn_id=102;
+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 |
| 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

Full TCL Example

```
MySQL 8.0 Command Line Cli × + ▾ - □ ×

1 row in set (0.00 sec)

mysql> SET autocommit=0;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
    -> SET amount=7000
    -> WHERE txn_id=101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
    -> SET amount=9000
    -> WHERE txn_id=102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM bank_transactions
    -> WHERE txn_id IN (101,102);
+-----+-----+-----+-----+-----+
-----+
```

```
mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
-> SET amount=9000
-> WHERE txn_id=102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM bank_transactions
-> WHERE txn_id IN (101,102);
+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount |
| transaction_date | account_no |
+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 7000.00 |
| 2024-01-05 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 |
| 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```